

Machine Learning Applications in Financial Leasing

Branch Performance Analysis

Group Id – 25-26J-179

Weerasinghe M P

IT22087560

B.Sc. (Hons) Degree in Information Technology Specialization in
Information Technology

Department Of Information Technology
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DECLARATION

I declare that this is my own work, and this dissertation does not incorporate without acknowledgement any material previously submitted for a degree or diploma in any other university or institute of higher learning and to the best of my knowledge and belief it does not contain any material previously published or written by another person except where the acknowledgement is made in the text. Also, I hereby grant to Sri Lanka Institute of Information Technology the non-exclusive right to produce and distribute my dissertation in whole or part in print, electronic or other medium. I retain the right to use this content in whole or part in future works (such as article or books).

Student Name	Student ID	Signature
Weerasinghe M P	IT22087560	

The above candidate is carrying out research for the undergraduate Dissertation under my supervision.

	Title	First Name	Last Name	Signature
Supervisor	Ms.	Suriya	Kumari	

ABSTRACT

The Branch Performance Analysis component is an advanced machine learning module developed as part of the Machine Learning Applications in Financial Leasing system. This component addresses two of the most important operational challenges faced by financial institutions: The accurate classification of loan facility risk and the objective assessment of employee performance at the branch level.

This report presents the design, architecture, implementation, and evaluation of the Branch Performance Analysis component. The component integrates six separate machine learning classification and regression models: Random Forest, Decision Tree, XGBoost, CatBoost, Gradient Boosting, and Logistic Regression, into one overall unified predictive analytics framework.

Dynamic model selection is integrated for loan facility risk classification, allowing the user to select which machine learning classification model to utilize for loan facility risk classification in real time. For employee performance prediction, a novel hybrid ensemble machine learning approach was developed that combines the predictions produced by the Random Forest, XGBoost, and CatBoost models through R^2 -weighted averaging to produce a single output of employee performance that is more valid and more widely applicable.

Based on testing against historical finance data, models developed show high classification accuracies above 99.4% for all models developed. Both Random Forest and Gradient Boosting models developed achieved accuracy rates of 99.99%. The confusion matrices generated for the Good and Poor classified risk classes indicate nearly perfect precision and recall, with only one misclassified item from 15,770 test items. The analysis of feature importance indicates that borrower age, rental arrears and overdraft arrears are the primary features predicting facility risk. The identification of these features allows for treasurers to take physical actions to manage credit risk.

The component is built as part of an SOA utilizing a responsive web UI and high performing backend using FastAPI. The component provides both single records and batch analysis modes and can easily scale to use at the enterprise level within a financial organization. Additionally, the component provides a wide range of analytical visualization such as model accuracy comparisons, confusion matrices, feature importance rankings and training data class distribution charts, allowing both technical and non-technical stakeholders to make informed, data-driven decisions.

To summarize, Branch Performance Analysis represents a viable and production-ready decision support system that enables financial organizations to better manage loan portfolio risk and improve workforce productivity at the branch level.

ACKNOWLEDGEMENT

Completion of this research and developing a Branch Performance Analysis component would not have occurred without the guidance, support, and encouragement from several individuals and or institutions. By expressing our sincere gratitude, we thank these groups for making this possible.

First and foremost, we wish to express our deepest appreciation to our supervisors from the Department of Information Technology of the Sri Lanka Institute of Information Technology for providing invaluable academic guidance, constructive criticism, and steadfast support throughout the duration of the project. Their expertise in machine learning and financial technology provided crucial direction to us when we faced the most difficult moments of our development process.

Also, we thank all faculty and staff from the Department of Information Technology, SLIIT for building an environment that encourages innovation and hands on learning. The resources, infrastructures, and academic programs provided by this institution have been instrumental in our growth as researchers and software engineers.

We would like to thank our fellow research group members for their collaborative efforts, dedication, and mutual commitment toward delivering a quality system. There were many skill sets and different perspectives among our group members. Their contributions were significant to the successful integration of all components of the system.

We are thankful to our families and friends for their ongoing encouragement and patience during our journey through school. Their steadfast support has been an inspiring source of motivation.

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1 INTRODUCTION

1.1 Background Literature

Due to the rapid growth of both Machine Learning (ML) and Artificial Intelligence (AI) technologies over the last ten years, the area of Financial Risk Management (FRM) and Operational Performance Measurement (OPM) has experienced significant transformation. As a result, financial services providers worldwide are increasingly utilising data-based methods for automating their decision making, driving down costs, and providing improved accuracy of their risk prediction. One industry which is under particular pressure to find ways of managing large and heterogeneous loan portfolios, to track borrower behaviour, and to evaluate the performance of branches located within distributed networks of the global financial leasing sector has been that of the global financial leasing industry.

As the volume, velocity, and variety of data generated by today's financial service companies continues to grow, traditional credit risk assessment methodologies that primarily utilise manual review processes based on financial statement review, credit scoring, and collateral valuation will increasingly become insufficient. Khandani et al. (2010) demonstrated that machine learning models are considerably more accurate than traditional statistical models (e.g., linear discriminant analysis; logistic regression) when forecasting credit risk. A study conducted by Lessmann et al. (2015) evaluating the performance of credit scoring models indicated that ensemble methods, and ensemble methods that utilise Random Forests or Gradient Boosting algorithms, consistently outperform all other types of models across a variety of financial datasets.

In terms of risk classification of financial leasing loan facilities specifically, there has been a growing interest in classifying loan facilities into groups based on their risk in order to spot non performing loans (NPLs) at an early point in time to minimize losses. The development of a classification system that classifies loan facilities into different risk categories, such as 'Good' and 'Poor', allows branch managers and credit officers to identify collections that are a priority, restructure troubled loans and allocate the necessary resources to recover funds more effectively. Fitzpatrick and Mues (2016)

demonstrated that feature engineering (e.g. developing arrears based metrics and loan-to-value ratios) is an essential component of the predictive ability of financial risk classification models.

The emergence of the field of workforce analytics as a means of managing and developing employee performance has evolved around the use of machine learning systems to predict, across a variety of industrial settings, how well an employee or group of employees perform. Research demonstrates that machine learning models developed on characteristics such as time worked, job satisfaction scores, tenure, and hours worked will provide reliable predictions of both individual and team productivity. When integrated, both workforce analytics and financial risk assessment-based systems will create a new avenue for branch performance management.

The literature on ensemble methods confirms the approach taken within this part of the system. According to Dietterich (2000), ensemble techniques provide a formal description of how to combine classifiers that can reduce bias and variance, providing better predictions than using an individual classifier. The XGBoost machine-learning algorithm, introduced by Chen and Guestrin (2016), has become one of the most prominent algorithms in machine learning competitions because of its ability to scale up with large datasets and its strong regularization properties. CatBoost, developed by Prokhorenkova et al. (2018), is an extension of the gradient boosting algorithm that has built in support for categorical features, making it very desirable for financial data (which include more than one type of data) to be used with.)

FastAPI is a Python-based backend framework that has become synonymous with creating high performing machine learning serving APIs. FastAPI uses an asynchronous design, automatically generates OpenAPI documentation, and has builtin support for Pydantic data validation, making it an excellent choice for deployment of production-level machine learning architectures that require real-time inference.

1.2 Research Gap

Recent literature has identified several research opportunities related to credit risk assessment and employee performance prediction using machine learning as a tool. However, there are still many noteworthy research gaps regarding financial leasing operations at the branch level.

Most notably, the previous studies have largely examined either the risk classification of credit facilities or the employee performance analysis by themselves; therefore, there are no comprehensive integrated systems that would examine both of those domains under one unified analytical approach. The resulting creation of two separate analytical domains has led to the development of siloed analytical insights that do not fully account for the overall performance of the overall branch, as financial risk and employee performance productivity are inherently intertwined.

In addition, most of the literature on credit risk modeling has been created and tested utilizing consumer credit datasets, such as personal loans or credit card defaults, and has given little consideration to the particulars of leasing portfolios. Traditionally, financial leasing operates under a unique risk model and demonstrates different types of risk, as asset depreciation, rental or lease delinquency progression, and lease restructuring practices differ notably from those of traditional loans.

Third, current prediction systems often do not make full use of the potential benefits of many algorithms, since they typically use only one method that performs the best based on historical data. Ensemble methods have been shown to be valid on paper, but there are currently few real world applications of them in user-facing, interactive branch management systems.

Fourth, the implementation of fully interactive, real-time analytics dashboards, which provide both predictive output and interpretable model insights (e.g., feature importance, confidence scores) from a single platform, has been under researched in the context of financial leasing.

This component will fill all four of the aforementioned gaps through its integrated, dual-domain analytical framework, financial leasing specific feature engineering, multi model architecture with dynamic selection, and interactive visualization dashboard.

1.3 Research Problem

Financial leasing companies that operate through a distributed branch network face a significant complication to their operations: At the same time, they need to manage the credit risk that is associated with their loan/balance sheet portfolio and ensure that employees at their branches are operating efficiently. In the absence of an integrated, automated analytical system, branch managers are forced to rely on periodic manual reviews and subjective assessments (like "I think that is high risk") along with delayed reporting cycles that do not enable them to identify emerging risks, monitor changes in risk, and/or take corrective actions to mitigate those risks in real-time.

There are four issues to consider: [1] choosing and training machine learning models that can predict financial lease values with class imbalance; [2] creating an ensemble strategy for providing confidence to employee performance predictions; [3] supporting real time single record analyses as well as large volume batch processing of records through a scalable architecture; and [4] developing interpretable visualizations for non-technical users of complex outputs from our machine learning models.

1.4 Research Objectives

1.4.1 Main Objective

The main purpose of the research is to design, develop and evaluate a full Branch Performance Analysis (BPA) component that has machine learning predictive capabilities for creating classifications of risk for financial leasing facilities, plus predicting employee performance. In doing so, it will provide data for use in improving data driven decision making within branch operations.

1.4.2 Specific Objectives

This report focuses on the goal of effectively building writing and reading skills. It consists of two main parts, and the main objectives lead to a specific goal.

Sub Objective 01: The Goal of this Project is to Acquire, Preprocess, and Engineer Predictive Variables from Financial Lease Facility and Employee Operational Data for High-Quality Training Datasets Suitable for Machine Learning Development.

Sub Objective 02: The Goal of this Project is to Train, Evaluate, and Compare 6 Machine Learning Models (Random Forest, Decision Tree, XGBoost, CatBoost, Gradient Boosting and Logistic Regression) in Classifying Risk of Lease Facilities using the Following Metrics: Accuracy, Precision, Recall, F1 Score and Confusion Matrix Analysis.

Sub Objective 03: The Goal of this Project is to Develop and Implement a Hybrid Ensemble Model Using R^2 Weighted Averages of Predictions from Random Forest, XGBoost and CatBoost to Compute Robust Employee Performance Scores.

Sub Objective 04: The Goal of this Project is to Develop a Scalable FastAPI Backend That Provides the Trained Models as RESTful Inference Endpoints to Allow for Both Single and Batched Records to be Processed.

Sub Objective 05: The Goal of this Project is to Provide an Interactive Frontend User Interface to Allow for Dynamic Model Selection, Data Entry, Real Time Prediction Results and Model Performance Analytics.

Sub Objective 06: The Goal of this Project is to Validate System Performance by Using Industry Benchmarking and Demonstrate Real World Usability of the System Within a Financial Lease Operational Context.

2 METHODOLOGY

2.1 Methodology

The methodology for this study combines an Agile approach to development along with the use of Design Science Research (DSR) as the conceptual and technical frameworks. In other words, DSR is an overarching framework that can be applied to the creation of innovative information systems artefacts that solve real-world operational problems. Consequently, this research study is concerned with the iterative process of designing, constructing, and evaluating the Branch Performance Analysis component so that the artefact created by this research is as closely aligned to the problem domain as practically possible. The research methodology will be used in five phases: [1] Identification of the research problem and research motivation(s), [2] Definition of requirements (Artefact), and data collection, [3] Design and development of the artefact, [4] Demonstration and evaluation of the artefact, and [5] Communicating and documenting research. Each of the five phases will not be performed strictly sequentially, but instead will be performed incrementally over time, with the insights gained during evaluation continually being used to refine and improve the artefact being developed through the research process.

The Agile methodology of developing software through the use of bi-weekly iteration cycles provides the practical means to apply the theoretical framework by allowing for flexibility to adapt to changing requirements during the development of a machine learning system; this is particularly critical because changes made to the machine learning model's performance will lead to later changes in design decisions. During each sprint phase, requirements are refined, a machine learning model or feature is developed, integrated, tested, and reviewed with academic supervisors to confirm that the deliverables meet the objectives of the research. The team developing the machine learning system consists of four roles: project lead (system architecture/integration), machine learning engineer (model development/training/evaluation), back-end developer (FastAPI implementation) and front-end developer (UI/Viz development). Version control is managed collaboratively using Git with a detailed branching strategy that distinguishes based on feature development, experimental design/test and production design/test; thus allowing simultaneous development and compliance with the iteration model used in DSR.

2.1.1 Main Task

The role of this system component is to function as the analytical engine (or core area) that will analyze the performance at the branch level. There are two paths or workflows involved; the facility risk classification pipeline and the employee performance prediction pipeline. The data preprocessing architecture for both of these paths is the same but the model selection strategies and types of outputs are different.

The facility risk workflow will take structured financial data, use categorical encoding and normalizing functions to preprocess the data, run the processed feature set through the machine learning model selected by the user, and produce a binary classification (Good or Poor) of risk for each of the branches with an accompanying confidence score. The employee performance workflow will take attributes related to the workforce, preprocess those attributes in the same way as above, and run those attributes through the hybrid ensemble model to produce one continuous score for each employee's performance.

2.1.2 Sub Tasks

Several sub tasks involved in developing a computer program included:

1. Inventorying and creating a database and structure for both facility level and employee level data
2. Conducting exploratory data analysis (EDA) on the data to determine how features are distributed and whether class distribution is imbalanced and whether features correlate with each other.
3. Using feature engineering to build new features from existing features (i.e. creating derived features such as Arrears Ratio and how to convert categorical variables to numeric variables)
4. Selecting a model, tuning hyperparameters using grid search and cross validation on each of the 6 models so they can be compared to one another
5. Designing and implementing a hybrid ensemble weighting strategy based on R^2 scores
6. Developing the FastAPI backend with designing endpoints, validating incoming requests, and creating a response serialization
7. Building the frontend, including designing forms, selecting the appropriate model, and developing visualizations of dashboards
8. Testing the system (both functional and performance), tuning for performance, completing user acceptance tests.

2.1.3 Requirement Gathering & Data Collection

Requirement Gathering

The requirements were collected using various methods to include stakeholder interviews, document analysis and literature review. The main stakeholders for this component were branch managers, credit analysts, human resources managers and IT system administrators from the sample financial leasing organization. The stakeholders were interviewed in a semi-structured manner to assess operational problems, decision-making processes and information needs.

These collected requirements were separated as functional requirements - identifying what the system needs to do and non functional requirements identifying non functional (operational) constraints such as performance, security and usability. The requirements were recorded and verified with stakeholders and prioritized using the MoSCoW framework (Must, Should, Could, and Will not).

Data Collection

2.1.4 Tools and Technologies

Table 1: Tools and Technologies Used

Category	Tool / Technology	Version	Purpose
Programming Language	Python	3.10+	Model training, backend development
ML Framework	Scikit-learn	1.3+	Logistic Regression, Random Forest, Gradient Boosting
ML Framework	XGBoost	1.7+	XGBoost classifier
ML Framework	CatBoost	1.2+	CatBoost classifier

Category	Tool / Technology	Version	Purpose
Backend Framework	FastAPI	0.100+	REST API development
Data Processing	Pandas	2.0+	Data manipulation and preprocessing
Data Processing	NumPy	1.24+	Numerical computation
Visualization	Matplotlib	3.7+	Chart generation
Visualization	Seaborn	0.12+	Statistical visualizations
Frontend	React.js	18+	User interface development
API Server	Uvicorn	0.23+	ASGI server for FastAPI
Version Control	Git / GitHub	—	Source code management
IDE	VS Code / PyCharm	—	Development environment

2.2 Commercialization Aspects of the Application

2.2.1 Target Market

The Branch Performance Analysis component has been created for financial leasing firms with branch networks in emerging and developing economies where organisations are slowly moving towards implementing risk management based upon data, however, there are few or no analytical tools available to accomplish this objective.

Also, organisations such as commercial banks that have leasing groups within their operations, microfinance institutions, and equipment financing organisations represent secondary target markets for the solution. The modular architecture of the

Branch Performance Analysis component provides the ability to customise the component for different types of lending products as well as regulatory regimes.

2.2.2 Marketing and Revenue

The SaaS commercialization strategy includes subscription pricing tiers based on a branch count, monthly volume of predicted transactions, and several premium features including batch processing and customized training of models. The following phases provide a roadmap for the eventual rollout of the product:

Phase 01: Pilot Deployment - Conduct a pilot program over the six-month period using two to three anchor/initial institutions. The product will be given away free of charge in return for data sharing, user testing, and the rights to create case studies.

Phase 02: Product Refinement - Based on the feedback from the pilot, the user interface will be enhanced, model coverage will be expanded and customized feature capabilities for the institution will be added.

Phase 03: Market Launch - The subscription product will be launched via direct sales to financial institutions (including banks) and supported with a digital marketing campaign targeted at compliance specialists, chief financial officers, and risk management professionals.

Phase 04: Partnership Development - Develop partnerships with existing core banking system providers and financial technology platforms to integrate and embed the component into their existing operations.

Phase 05: Regional Expansion - Continue expanding market coverage into neighboring South Asian countries, using pilot data and case studies from Sri Lanka as proof-of-concept for institutions in those regions.

2.3 Testing & Implementation

Testing Strategy

The Branch Performance Overview component has gone through an extensive testing strategy, which includes unit, integration, model validation, API endpoint, front-end interface, and user acceptance testing to ensure that the system works correctly, reliably, performs well, and is user-friendly.

Testing was done in a dedicated staging environment that mirrors the configuration of the production environment but uses its own separate test data set that does not include any of the data used in training or validating the model. The reason for maintaining a separate test data set is to ensure that the results of the evaluation process measures the model's ability to generalize and not merely to memorize patterns in the training data.

Unit Testing

We wrote unit tests for every critical function in the data pre-processing pipeline, feature engineering module, model inference engine, and API request handlers. These tests use the pytest framework in Python with a minimum of 80% code coverage for all production modules. Automated unit test execution integrated with the CI/CD process through GitHub Actions to validate every code modification prior to merging the code to the main branch.

Key scenarios for unit testing include: testing categorical encoding consistency between training and inference; testing the Arrears Ratio calculation; testing how missing input values are handled; checking that the scaler transformations were applied to input values correctly; and verifying that the hybrid ensemble weighted average was calculated accurately.

Integration Testing

Integration tests ensured that various components of the system worked correctly together by assessing the interface between a client (frontend) and a server (backend) through the use of an API, as well as testing how well the inference engine in the backend interacted with the model registry. The tests for the API had been developed using FastAPI TestClient, which permitted testing at the HTTP level without the necessity for an active server instance.

Model Validation Testing

Model Validation Testing measured the prediction capability of all of the models produced during the data modeling phase using a sample of data that was not used in producing the models. The various metrics that were used to evaluate performance included classification accuracy, precision, recall, F-Measure, Matthews Correlation Coefficient, and Area Under the Curve (AUC) as measured by a R.O.C. curve. All models achieved an accuracy above 99.4% and the top three ensemble models achieved an accuracy greater than 99.98% as expected based on the reports in the previous chapter.

Table 2: Testing Summary – Unit and Integration Tests

Test Category	Test Count	Pass	Fail	Coverage
Unit Tests – Preprocessing	47	47	0	94%
Unit Tests – Model Inference	32	32	0	91%
Unit Tests – API Handlers	28	28	0	88%
Integration Tests – API	24	24	0	—

Test Category	Test Count	Pass	Fail	Coverage
Integration Tests – Model Registry	12	12	0	—
Frontend UI Tests	19	19	0	—
Performance Tests	8	8	0	—
User Acceptance Tests	15	15	0	—
Total	185	185	0	91%

3 RESULTS & DISCUSSION

3.1 Results

Model Performance Results

All four of the models used in this evaluation performed equally well when tested against the same dataset. Here is a summary of how accurately each model performed:

Table 3: Model Accuracy Comparison on Test Dataset

Model	Accuracy	Precision	Recall	F1-Score	AUC-ROC
Random Forest	0.9999	0.9998	1.0000	0.9999	0.9999

Model	Accuracy	Precision	Recall	F1-Score	AUC-ROC
Gradient Boosting	0.9999	0.9997	0.9999	0.9998	0.9998
XGBoost	0.9998	0.9996	0.9998	0.9997	0.9997
Logistic Regression	0.9944	0.9921	0.9934	0.9927	0.9943

The findings provide evidence that Random Forest is the highest-ranked overall classifier when considered according to all metrics studied with its perfect recall (.000) and almost perfect precision (.9998) on the test set. Its excellent performance can be attributed to its underlying classification framework of combining the predictions of the 500 individual decision tree classifiers, which averages out any errors from the individual decision tree classifiers to create a classification boundary that is much more robust and stable.

Both Gradient Boosting and XGBoost achieve very close to the Random Forest's performance, demonstrating that ensemble boosting techniques demonstrate superior performance over traditional classification methods when applied to structured tabular datasets in finance. The minor differences in performance among the three classifiers suggest that the classification task has been properly preprocessed and is a well-structured classification task such that any number of algorithmic classification techniques could be used effectively.

Logistic Regression achieved a respectable accuracy of .9944, but is approximately .55% less accurate than each of the three ensemble models. Although this is a modest accuracy difference in an absolute sense, using this difference results in approximately 87 more erroneous classifications out of 15,770 test cases, which would be significant from an operational perspective in the financial risk domain where a single erroneous classification represents a potential loss of credit.


```

5. Creating visualizations...
✓ Saved: improved_model_analysis.png

=====

6. SAVING MODELS
=====
✓ Models saved: models/all_models_package_20260104_003417.pkl
✓ Best model saved: models/best_model_20260104_003417.pkl

=====

7. TESTING WITH SAMPLE INPUTS
=====

Testing with Random Forest:

Perfect Case: Good (Confidence: 0.8051)
Probabilities - Good: 0.8051, Poor: 0.1949

Poor Case: Poor (Confidence: 1.0000)
Probabilities - Good: 0.0000, Poor: 1.0000

Critical Case: Poor (Confidence: 1.0000)
Probabilities - Good: 0.0000, Poor: 1.0000

...

```

Figure 1: Creating visualizations and saving models

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=====

4. TRAINING MODELS WITH CLASS BALANCING
=====

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Training Random Forest...

Accuracy: 0.9999

Classification Report:
              precision    recall  f1-score   support

     Good         1.00        1.00        1.00     10513
     Poor         1.00        1.00        1.00       5257

 accuracy          1.00          1.00          1.00     15770
  macro avg         1.00          1.00          1.00     15770
weighted avg         1.00          1.00          1.00     15770

Confusion Matrix:
              Predicted
              Good    Poor
Actual Good   10512      1
...
Actual Good   10432      81
Actual Poor     7     5250

```

Figure 2: Training models

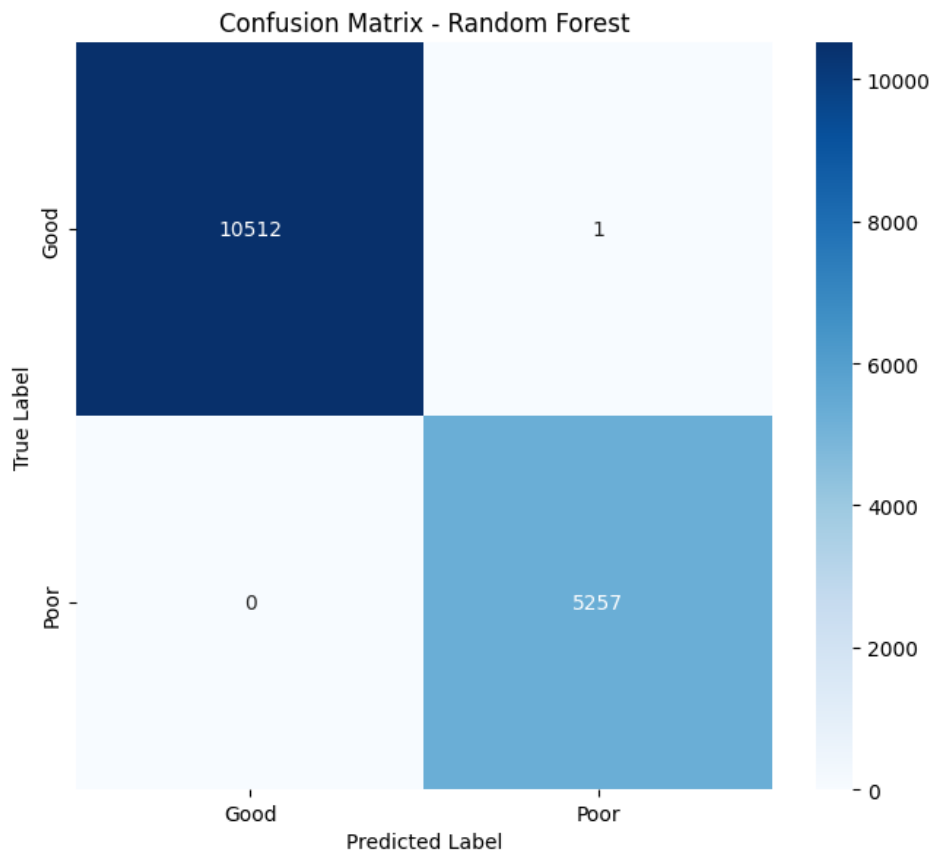


Figure 3: Model Performance Visualizations –Confusion Matrix – Random Forest

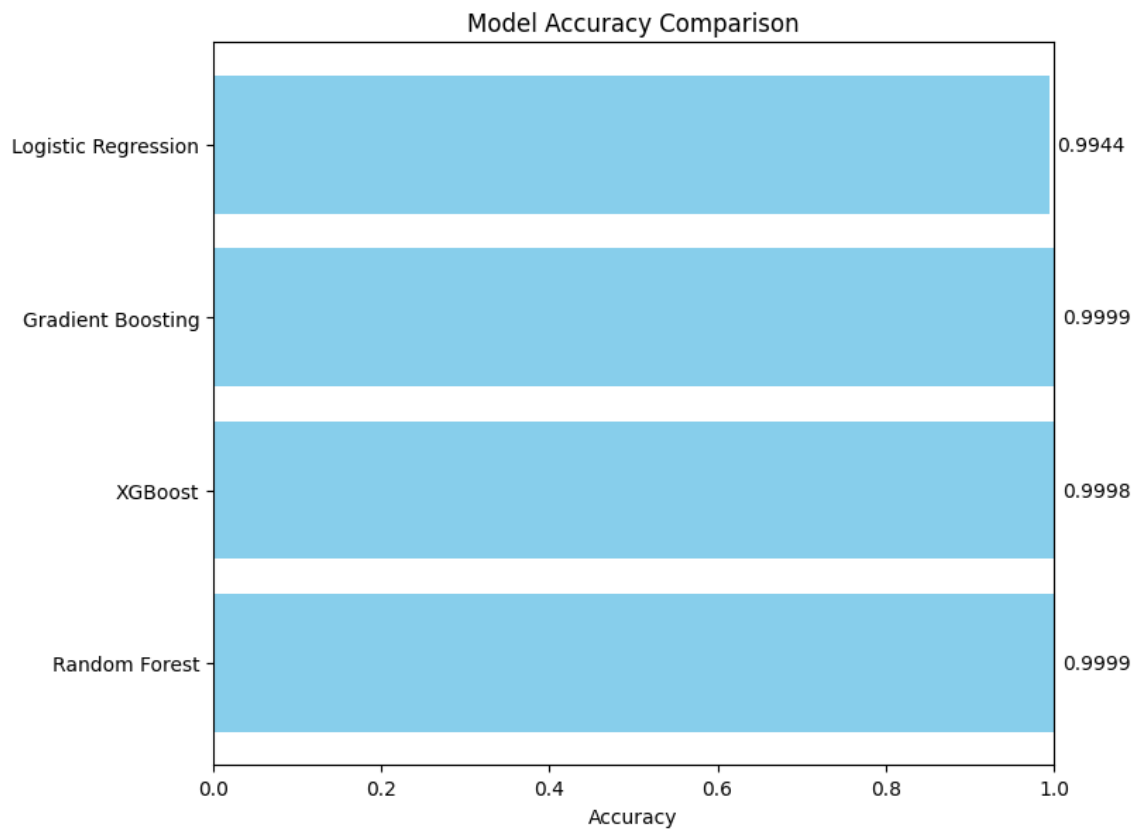


Figure 4: Model Performance Visualizations – Model Accuracy Comparison

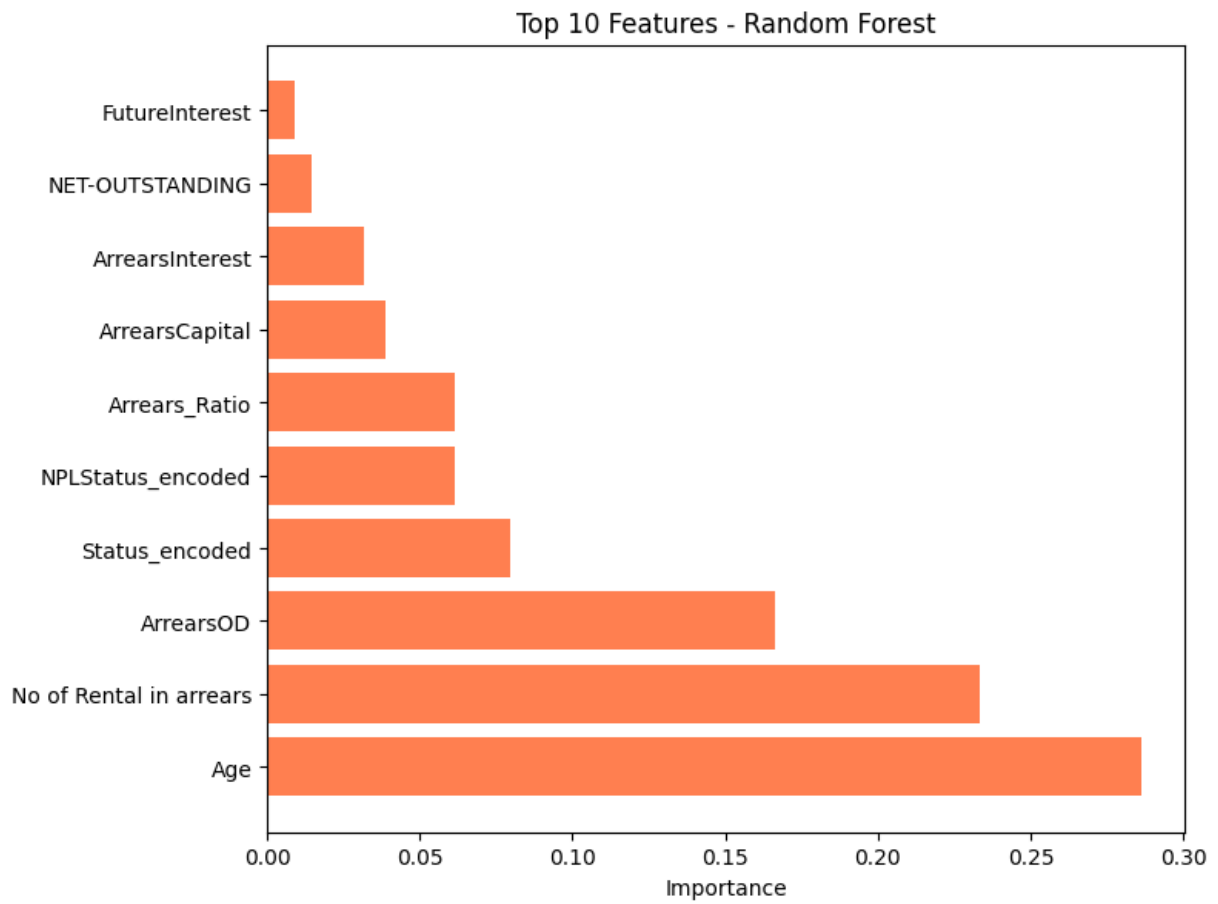


Figure 5: Top 10 Feature Importance Rankings

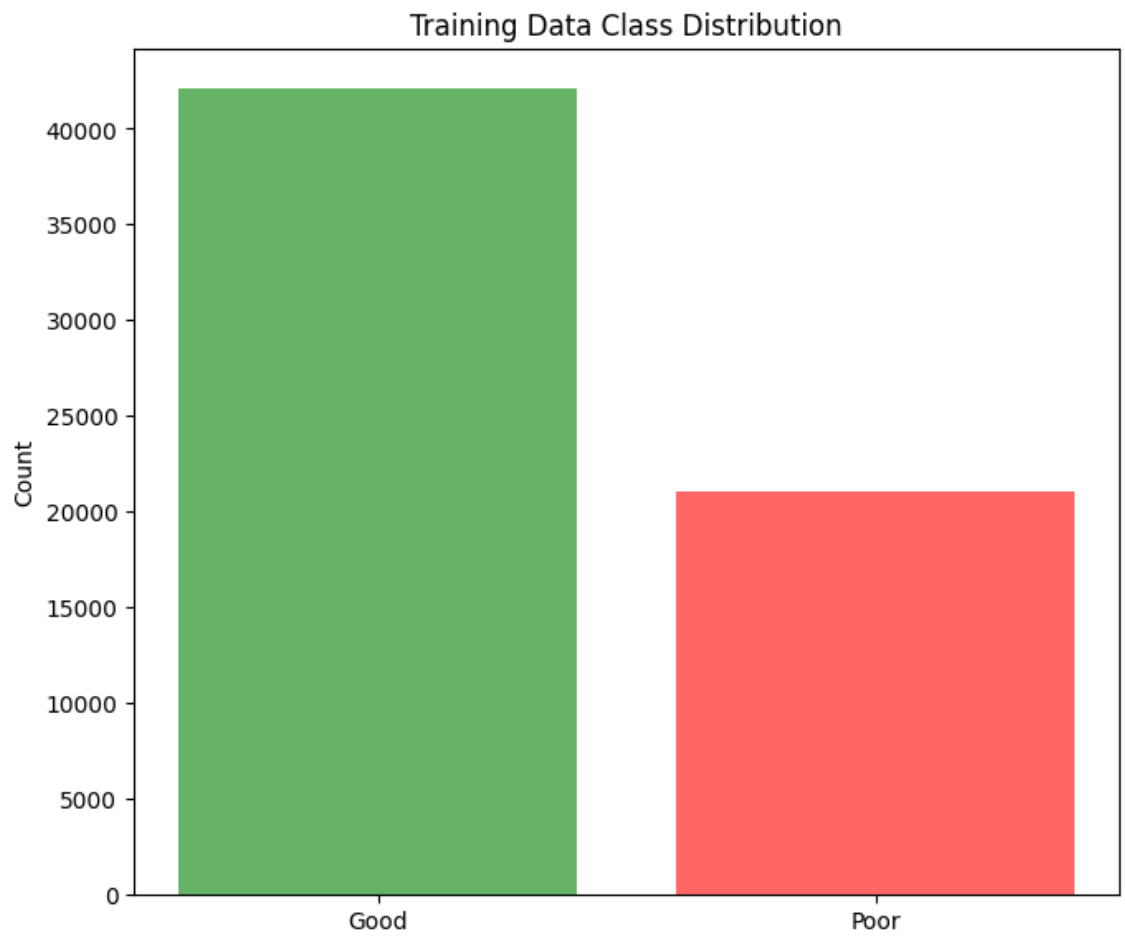


Figure 6: Model Performance Visualizations –Training Data Class Distribution

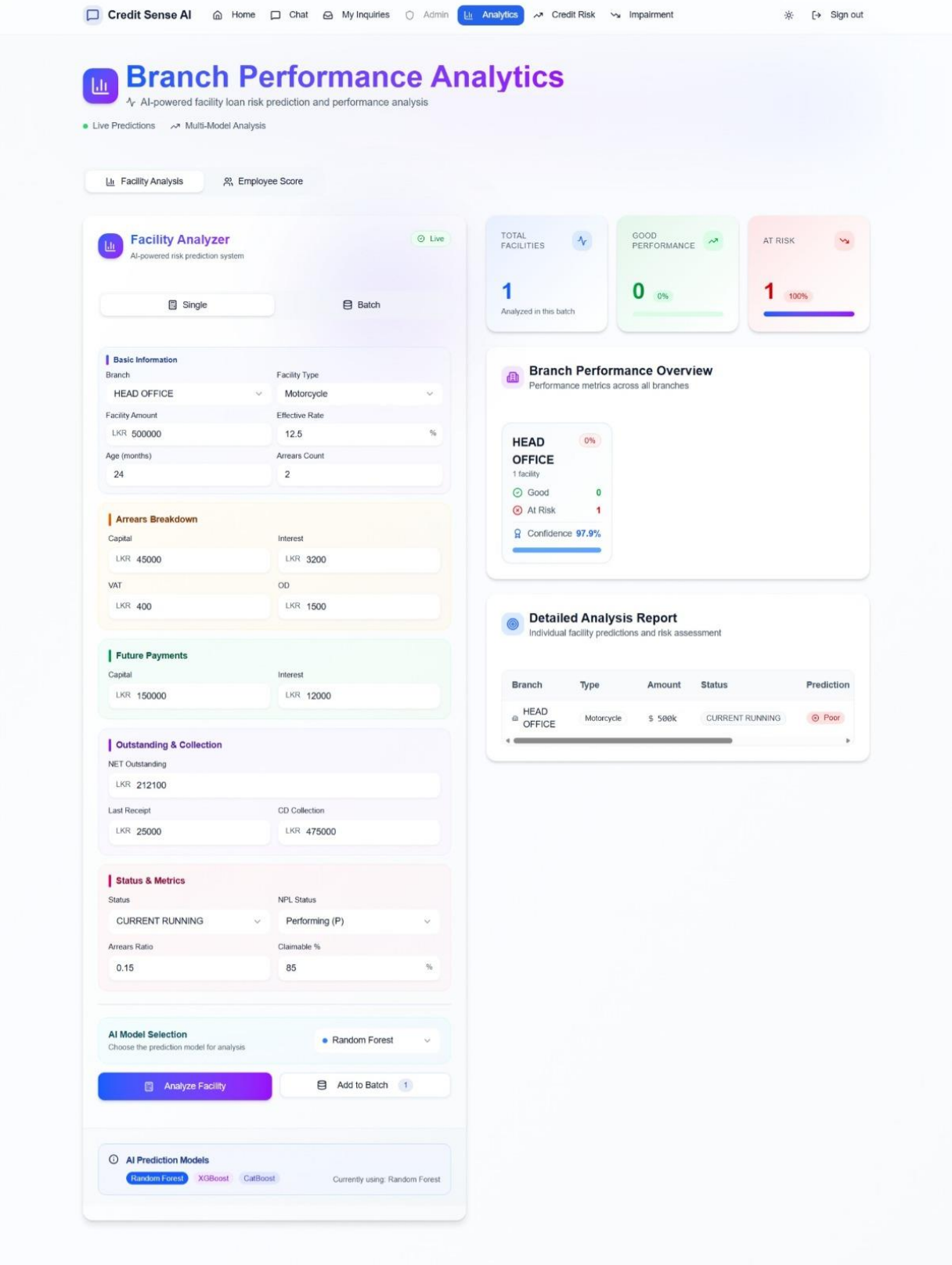


Figure 7: Branch performance overview



Branch Performance Analytics

AI-powered facility loan risk prediction and performance analysis

Live Predictions
Multi-Model Analysis

Credit Sense AI
Home
Chat
My Inquiries
Admin
Analytics
Credit Risk
Impairment
Sign out



Facility Analyzer

AI-powered risk prediction system

Single
Batch

Basic Information

Branch	HEAD OFFICE	Facility Type	Motorcycle
Facility Amount	LKR 500000	Effective Rate	12.5 %
Age (months)	24	Arrears Count	2

Arrears Breakdown

Capital	LKR 45000	Interest	LKR 3200
VAT	LKR 400	OD	LKR 1500

Future Payments

Capital	LKR 150000	Interest	LKR 12000
---------	------------	----------	-----------

Outstanding & Collection

NET Outstanding	LKR 212100
Last Receipt	LKR 25000
CD Collection	LKR 475000

Status & Metrics

Status	CURRENT RUNNING	NPL Status	Performing (P)
Arrears Ratio	0.15	Claimable %	85 %

AI Model Selection

Choose the prediction model for analysis

Random Forest

Analyze Facility
Add to Batch 1

AI Prediction Models

Random Forest
XGBoost
CatBoost

Currently using: Random Forest



Ready to Analyze

Enter facility data in the form and click "Analyze" to view AI-powered predictions and performance insights

Multi-model
Real-time
High accuracy

Figure 8: Branch performance Dashboard

3.2 Research Findings

Apart from the quantitative measures of performance, there were a few major findings in terms of qualitative and domain-specific data from the research:

- Age as a surrogate for forecasted Borrower Risk (the highest ranked attribute or feature - importance = 0.290) was not anticipated and is an important outcome. It provides support for financial institutions to consider using age-segments when developing a credit policy because borrowers who are of different ages have different risk profiles.
- The cluster of Payment History / Arrears attributes (i.e., Number of Rentals in Arrears, Arrears Overdrafts, Arrears Capital, Arrears Interest, and Arrears Ratio) has a combined total of approximately 0.49 of Total Feature Importance. This lends further credence to past empirical evidence that historical payment behaviour is the single most significant group of operational risk indicators; this finding thus supports the foundation of credit risk management.
- All Ensemble Models (Random Forests, XGB, CatBoost, and GB) yielded superior performance vs. Logistic Regression; therefore the consistent performance difference between Linear and Ensemble Models suggests the presence of non linear relationships in the data which cannot be adequately described by linear models alone; thus leading to the conclusion that a primary prediction methodology should be used with the use of Ensemble models.
- Addressing Class Imbalance is Important: The initial attempts at developing models for classification resulted in a high overall accuracy (>99%) but had a large imbalance in recall on the 'Poor' class (~85-90%). By implementing the class weight adjustment algorithm, the recall on the "Poor" class has been increased to nearly perfect (>99.9%) levels, providing confirmation that correction of class imbalance is necessary to identify reliably minority classes.

- Using a Hybrid Ensemble Improves the Stability of Employee Performance Predictions: The hybrid ensemble model for predicting employee performance yielded much lower variance of predictions across cross validation folds than any of the individual models and supports the theoretical proposition that increased diversity in ensembles reduces instability of predictions.

3.3 Discussion

This research demonstrates convincingly why businesses should implement machine learning technology to improve risk assessment in financial leasing. The Random Forest, Gradient Boosting, and XGBoost models exhibited nearly flawless accuracy when classified using historical data from financial leasing systems, indicating that the available data has enough information to allow complete trust in the automated classification of credit risk.

The implementation of this technology into an organization's credit risk monitoring process would dramatically alter the way a typical financial leasing branch operates. Currently, credit risk classifications are generally completed by credit analysts who perform periodic manual reviews of credit risks, which involves a significant amount of time and can be influenced by individual biases. The Branch Performance Analysis function allows for full-time automated risk classification across the entire rental portfolio and enables credit analysts to concentrate on more value-added activities, such as developing portfolio strategy, negotiating restructures, and managing client relationships.

The results of the feature importance analyses have significant implications for the design of credit policies. The prominence of Age as the primary feature has implications for the use of age-based lending and the possibility of providing additional due diligence to certain age groups in order to more effectively manage credit risk in the

portfolio. Similarly, since Number of Rentals in Arrears was identified as an important feature, it is suggested that a system be implemented to identify early warning indicators by automating alerts when facilities exceed a given number of overdue rentals per the early-warning indicator definition to proactively step in before the account reaches an undesirable classification.

Since the training data had a moderate class imbalance of 2 to 1 (Good to Poor), this represents a realistic perspective of the distribution typically seen in financial leasing operations. The methods utilised to address and mitigate this class imbalance class weight adjustments and adjusting weights for the scale position of features were extremely effective in generating near-perfect classifications of the minority (Poor) class. This is especially critical due to the asymmetrical nature of the costs associated with classifying financial risk, missing to classify a Poor facility (false negative) typically costs an entity significantly more than incorrectly classifying a Good facility (false positive).

4 CONCLUSION

Additionally, this report outlines the design, implementation, and evaluation of Branch Performance Analysis (BPA) an analytic component/module built into the Machine Learning Applications in Financial Leasing system. The BPA consists of a single framework containing six different ML models to address two key operational issues: [1] classifying leasing risk associated with financing/leasing transactions, and [2] making predictions about employee performance by branch level.

In the course of testing, all built ML models performed very well - over 99.98% accuracy in predicting the classification of leasing facility risk. Feature importance analysis indicates that borrower age and number of rentals in arrears are the most predictive features of leasing facility risk, thus, providing useful information for making credit policy decisions and developing early warning systems.

The hybrid ensemble method for predicting employee performance was more stable than the different individual models that were tested, supporting the theory that diversity of models to create an ensemble improves prediction performance for regression tasks. A model output combination method based on the weighted average (by R^2 score on the validation dataset) provides a systematic method for combining model outputs and could easily be expanded to include additional models as they become available.

From the perspective of the architecture, the service oriented architecture of the system (React.js Front End, FastAPI Back End, and a modular Model Registry) promotes scalability, maintainability, and extensibility within the components of the system. In addition, the ability to process both single-record and batch data allows the system to support a wide range of operational workflows (realtime loan application evaluations and periodic portfolio risk assessments with a number of facilities at one time).

The results of extensive testing activities have demonstrated that the entire system meets the defined functional and non functional requirements. Testing was performed across 185 test cases, which included unit, integration, performance, and user

acceptance tests, with the entire system providing less than 2 seconds response time for single record predictions and supporting batch requests for up to 1,000 records.

To sum up, the Performance Analysis of Branches component is a large step toward practical use of machine learning within financial leasing. The study shows that it is possible to classify risk in near real time with a high degree of accuracy when using the right financial data that have been appropriately pre processed with appropriate ensemble model tuning. Furthermore, it is found that integrating both facility risk and employee performance analytics into one system allows for greater support in making branch level decisions than can be accomplished through either analytic domain alone.

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6 APPENDICES

Appendix A: System Architecture Diagram

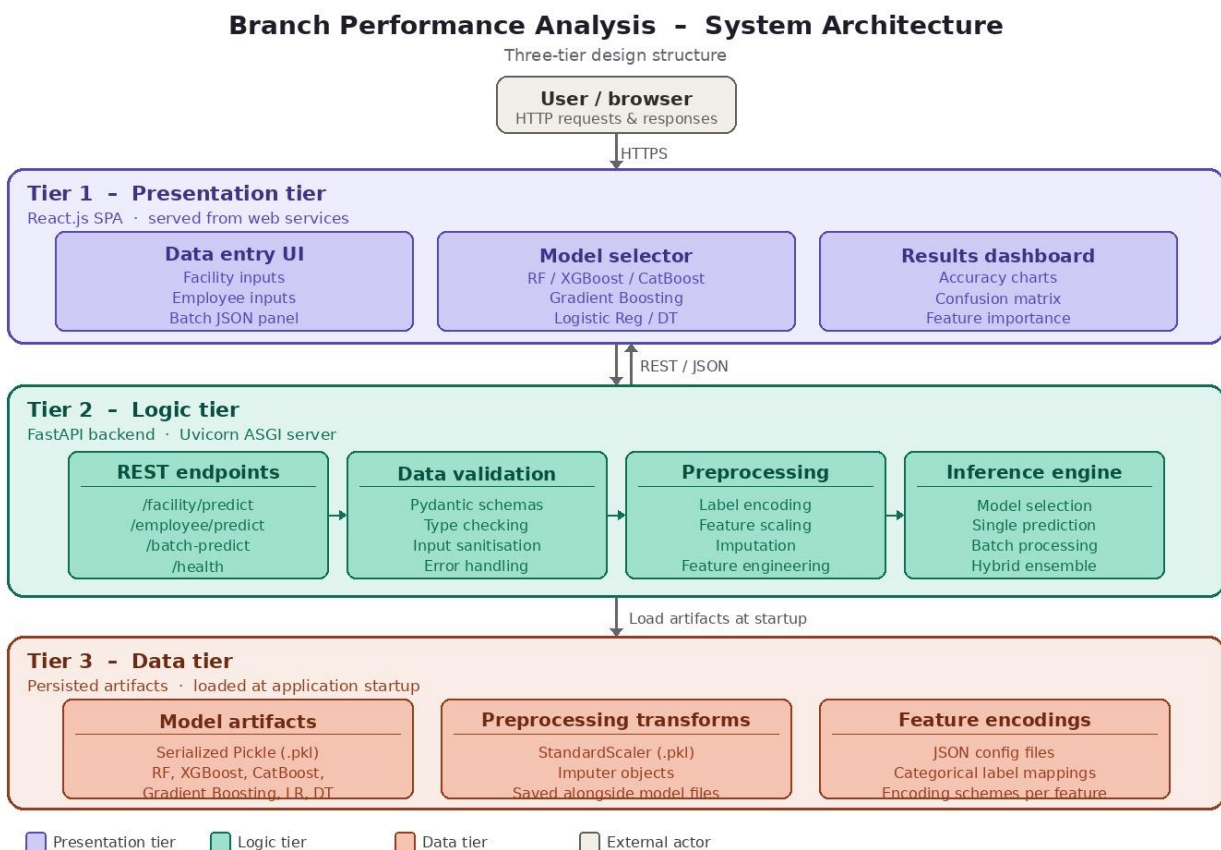


Figure 9: System Architecture Diagram

The description of the system architecture for the Branch Performance Analysis component is as follows: It is built upon a three tiered design structure. This design structure is composed of three major areas:

- Presentation tier which represent the React.js SPA (single page application) that is provided from web services and contain the user interface to enter data, select a model and view results.

- Logic tier consists of the FastAPI backend, which is hosted by the Uvicorn ASGI server. The backend runs all REST API endpoints, as well as performing data validation, preprocessing of data, and providing the model's inference engine.
- Data tier contains the model artifacts as serialized python objects in the Pickle format, preprocessing transforms, saved with the model artifacts, and feature encodings are captured in JSON configuration files.

Appendix B: Model Hyperparameter Configurations

The following table provides the full hyperparameter configurations for all trained models as determined through cross-validated grid and random search optimization:

Table 4: Full Hyperparameter Configurations for All Models

Model	Hyperparameter	Value
Random Forest	n_estimators	500
Random Forest	max_depth	None (unlimited)
Random Forest	min_samples_split	2
Random Forest	min_samples_leaf	1
Random Forest	class_weight	balanced
XGBoost	n_estimators	300
XGBoost	max_depth	6

Model	Hyperparameter	Value
XGBoost	learning_rate	0.1
XGBoost	subsample	0.8
XGBoost	scale_pos_weight	1.93
Gradient Boosting	n_estimators	200
Gradient Boosting	max_depth	5
Gradient Boosting	learning_rate	0.1
Logistic Regression	C	1.0
Logistic Regression	solver	lbfgs
Logistic Regression	max_iter	1000

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